

Results

July 1, 2026

1 Tables of Friedman, Bonferroni-Dunn, Holm, Hochberg and Hommel Tests

Table 1: Average Rankings of the algorithms

Algorithm	Ranking
METEOR	5.6860
ROUGE-L	6.5960
BLEU-4	6.9290
BERTScore	6.4900
IRES	3.8720
SentenceBERT	4.8370
USE	5.9090
InferSent	7.3930
BLEURT	5.6940
SIDE	6.5420
SimLLM	6.0520

Friedman statistic considering reduction performance (distributed according to chi-square with 10 degrees of freedom: 444.39727272712884.
P-value computed by Friedman Test: 2.0454593574470437E-10.

Iman and Davenport statistic considering reduction performance (distributed according to F-distribution with 10 and 4990 degrees of freedom: 48.677255758775615.

P-value computed by Iman and Davenport Test: 1.4252813512799767E-93.

Table 2: Holm / Hochberg Table for $\alpha = 0.05$

i	algorithm	$z = (R_0 - R_i) / \sqrt{SE}$	p	Holm/Hochberg/Hommel
10	InferSent	16.785708883668686	3.10495024128874E-63	0.005
9	BLEU-4	14.573675676618908	4.130550871277846E-48	0.005555555555555556
8	ROUGE-L	12.986160465524987	1.4659915078915544E-38	0.00625
7	SIDE	12.728725566428727	4.094681419615857E-37	0.0071428571428571435
6	BERTScore	12.480825293224854	9.499685886995614E-36	0.008333333333333333
5	SimLLM	10.392742222776953	2.675481714965443E-25	0.01
4	USE	9.71101647146638	2.7062466592891625E-22	0.0125
3	BLEURT	8.686044188027378	3.7527897892724421E-18	0.016666666666666666
2	METEOR	8.647905684457566	5.245315319704807E-18	0.025
1	SentenceBERT	4.600456993109997	4.215650541578348E-6	0.05

Bonferroni-Dunn's procedure rejects those hypotheses that have a p-value ≤ 0.005 .
Hochberg's procedure rejects those hypotheses that have a p-value ≤ 0.05 .
Hommel's procedure rejects all hypotheses.

Table 3: Holm / Hochberg Table for $\alpha = 0.10$

i	algorithm	$z = (R_0 - R_i) / \sqrt{SE}$	p	Holm/Hochberg/Hommel
10	InferSent	16.785708883668686	3.10495024128874E-63	0.01
9	BLEU-4	14.573675676618908	4.130550871277846E-48	0.011111111111111112
8	ROUGE-L	12.986160465524987	1.4659915078915544E-38	0.0125
7	SIDE	12.728725566428727	4.094681419615857E-37	0.014285714285714287
6	BERTScore	12.480825293224854	9.499685886995614E-36	0.016666666666666666
5	SimLLM	10.392742222776953	2.675481714965443E-25	0.02
4	USE	9.71101647146638	2.7062466592891625E-22	0.025
3	BLEURT	8.686044188027378	3.7527897892724421E-18	0.03333333333333333
2	METEOR	8.647905684457566	5.245315319704807E-18	0.05
1	SentenceBERT	4.600456993109997	4.215650541578348E-6	0.1

Bonferroni-Dunn’s procedure rejects those hypotheses that have a p-value ≤ 0.01 .
Hochberg’s procedure rejects those hypotheses that have a p-value ≤ 0.1 .
Hommel’s procedure rejects all hypotheses.

Table 4: Adjusted p -values

i	algorithm	unadjusted p	p_{Bonf}	p_{Holm}	p_{Hoch}	p_{Hommel}
1	InferSent	3.104995024128874E-63	3.1049950241288737E-62	3.1049950241288737E-62	3.1049950241288737E-62	3.1049950241288737E-62
2	BLEU-4	4.130550871277846E-48	4.1305508712778463E-47	3.71749578415000610E-47	3.71749578415000610E-47	3.71749578415000610E-47
3	ROUGE-L	1.4659915078915544E-38	1.4659915078915543E-37	1.1727932063132433E-37	1.1727932063132433E-37	1.1727932063132433E-37
4	SIDE	4.094681419615857E-37	4.094681419615857E-36	2.8662769937310997E-36	2.8662769937310997E-36	2.8662769937310997E-36
5	BERTScore	9.49968586995614E-36	9.49968586995614E-35	5.699811532197369E-35	5.699811532197369E-35	5.699811532197369E-35
6	SimLLM	2.675481714965443E-25	2.6754817149654428E-24	1.3377408574827214E-24	1.3377408574827214E-24	1.3377408574827214E-24
7	USE	2.7062466392891625E-22	2.7062466392891625E-21	1.082498663715665E-21	1.082498663715665E-21	1.082498663715665E-21
8	BLEURT	3.752789782724421E-18	3.7527897827244204E-17	1.1258369348173262E-17	1.0490630639409615E-17	7.867972979557211E-18
9	METEOR	5.245315319704807E-17	5.245315319704807E-17	1.1258369348173262E-17	1.0490630639409615E-17	1.0490630639409615E-17
10	SentenceBERT	4.215650541578348E-6	4.2156505415783485E-5	4.215650541578348E-6	4.215650541578348E-6	4.215650541578348E-6

Nemenyi’s procedure rejects those hypotheses that have a p-value $\leq 9.0909090909091E-4$.
Holm’s procedure rejects those hypotheses that have a p-value $\leq 0.0029411764705882353$.
Shaffer’s procedure rejects those hypotheses that have a p-value $\leq 9.0909090909091E-4$.
Nemenyi’s procedure rejects those hypotheses that have a p-value ≤ 0.00181818181818182 .
Holm’s procedure rejects those hypotheses that have a p-value ≤ 0.00625 .
Shaffer’s procedure rejects those hypotheses that have a p-value ≤ 0.00181818181818182 .

Table 5: Holm / Shaffer Table for $\alpha = 0.05$

i	algorithms	$z = (R_0 - R_i)/SE$	p	Holm	Shaffer
55	IRIS vs. InferSent	16.785708883668686	3.104995024128874E-63	9.090909090909091E-4	9.090909090909091E-4
54	BLEU-4 vs. IRIS	14.573675676618908	4.130550871277846E-48	9.25925925925926E-4	0.0011111111111111111
53	ROUGE-L vs. IRIS	12.986160465524987	1.4659915078915544E-38	9.433962264150943E-4	0.0011111111111111111
52	IRIS vs. SIDE	12.728725566428727	4.094681419615857E-37	9.615384615384616E-4	0.0011111111111111111
51	BERTScore vs. IRIS	12.480825293224854	9.499685886995614E-36	9.80392156862745E-4	0.0011111111111111111
50	SentenceBERT vs. InferSent	12.18525189055869	3.725006267054306E-34	0.001	0.0011111111111111111
49	IRIS vs. SimLLM	10.392742227776953	2.675481714965443E-25	0.0010204081632653062	0.0011111111111111111
48	BLEU-4 vs. SentenceBERT	9.973218683508911	1.9965170154146324E-23	0.0010416666666666667	0.0011111111111111111
47	IRIS vs. USE	9.71101647146638	2.7062466592891625E-22	0.0010638297872340426	0.0011111111111111111
46	IRIS vs. BLEURT	8.686044188027378	3.752789782724421E-18	0.0010869565217391304	0.0011111111111111111
45	METEOR vs. IRIS	8.647905684457566	5.245315319704807E-18	0.0011111111111111111	0.0011111111111111111
44	ROUGE-L vs. SentenceBERT	8.38570347241499	5.0423294553438346E-17	0.0011363636363636365	0.0013513513513513514
43	METEOR vs. InferSent	8.13780319921112	4.0251441541170054E-16	0.0011627906976744186	0.0013513513513513514
42	SentenceBERT vs. SIDE	8.12826857331873	4.3546554116534997E-16	0.0011904761904761906	0.0013513513513513514
41	InferSent vs. BLEURT	8.09966469564131	5.511087099509436E-16	0.0012195121951219512	0.0013513513513513514
40	BERTScore vs. SentenceBERT	7.880368300114856	3.264175041129758E-15	0.00125	0.0013513513513513514
39	USE vs. InferSent	7.074692412202308	1.497806095635279E-12	0.001282051282051282	0.0013513513513513514
38	InferSent vs. SimLLM	6.3929666608917355	1.6269774607603165E-10	0.0013157894736842105	0.0013513513513513514
37	METEOR vs. BLEU-4	5.9257699921613405	3.1083741177045455E-9	0.0013513513513513514	0.0013513513513513514
36	BLEU-4 vs. BLEURT	5.88763148859153	3.917694549050277E-9	0.001388888888888889	0.001388888888888889
35	SentenceBERT vs. SimLLM	5.792285229666955	6.943508882068498E-9	0.0014285714285714286	0.0016129032258064516
34	SentenceBERT vs. USE	5.110559478356383	3.2120616484214425E-7	0.0014705882352941176	0.0016129032258064516
33	BLEU-4 vs. USE	4.862659205152527	1.158191566162712E-6	0.0015151515151515152	0.0016129032258064516
32	IRIS vs. SentenceBERT	4.600456993109997	4.215650541578348E-6	0.0015625	0.0016129032258064516
31	METEOR vs. ROUGE-L	4.338254781067421	1.4361860779334917E-5	0.0016129032258064516	0.0016129032258064516
30	BERTScore vs. InferSent	4.3048835904438345	1.6707356546645684E-5	0.0016666666666666668	0.001724137931034483
29	ROUGE-L vs. BLEURT	4.300116277497609	1.7070851629526506E-5	0.001724137931034483	0.001724137931034483
28	BLEU-4 vs. SimLLM	4.180933453841955	2.9031482616797085E-5	0.0017857142857142859	0.0017857142857142859
27	SentenceBERT vs. BLEURT	4.085587194917381	4.39654854323836E-5	0.001851851851851852	0.001851851851851852
26	METEOR vs. SIDE	4.08081988197116	4.487712190045435E-5	0.0019230769230769232	0.002
25	InferSent vs. SIDE	4.056983317239961	4.9710631289064145E-5	0.002	0.002
24	METEOR vs. SentenceBERT	4.04744869134757	5.177892726152464E-5	0.0020833333333333333	0.0020833333333333333
23	BLEURT vs. SIDE	4.0426813784013484	5.2843406528614187E-5	0.002173913043478261	0.002173913043478261
22	METEOR vs. BERTScore	3.8329196087672863	1.266313451372739E-4	0.002272727272727273	0.002272727272727273
21	ROUGE-L vs. InferSent	3.7995484181437007	1.4495999007542706E-4	0.002380952380952381	0.002380952380952381
20	BERTScore vs. BLEURT	3.7947811051974756	1.477737716707003E-4	0.0025	0.0025
19	ROUGE-L vs. USE	3.2751439940586065	0.0010560812899933075	0.002631578947368421	0.002631578947368421
18	USE vs. SIDE	3.0177090949623464	0.002546932671489676	0.0027777777777777778	0.0027777777777777778
17	BERTScore vs. USE	2.769808821758473	0.0056089203911152	0.0029411764705882353	0.0029411764705882353
16	ROUGE-L vs. SimLLM	2.5934182427480343	0.009502712976774248	0.003125	0.003125
15	SIDE vs. SimLLM	2.3359833436517743	0.019492109652479973	0.0033333333333333333	0.0033333333333333333
14	BLEU-4 vs. InferSent	2.2120332070497803	0.026964370555333163	0.0035714285714285718	0.0035714285714285718
13	BLEU-4 vs. BERTScore	2.092850383394054	0.03636251257329595	0.0038461538461538464	0.0038461538461538464
12	BERTScore vs. SimLLM	2.088083070447901	0.03679034191083815	0.0041666666666666667	0.0041666666666666667
11	BLEU-4 vs. SIDE	1.8449501101901808	0.0650447962827122	0.004545454545454546	0.004545454545454546
10	METEOR vs. SimLLM	1.7448365383193858	0.08101332662327833	0.005	0.005
9	BLEURT vs. SimLLM	1.7066980347495746	0.08787819269001575	0.005555555555555556	0.005555555555555556
8	ROUGE-L vs. BLEU-4	1.5875152110939206	0.11239600465412686	0.00625	0.00625
7	METEOR vs. USE	1.0631107870088137	0.2877317131831188	0.0071428571428571435	0.0071428571428571435
6	USE vs. BLEURT	1.0249722834390025	0.3053762657842809	0.008333333333333333	0.008333333333333333
5	USE vs. SimLLM	0.6817257513105721	0.49541238110767266	0.01	0.01
4	ROUGE-L vs. BERTScore	0.5053351723001337	0.6133234430319932	0.0125	0.0125
3	ROUGE-L vs. SIDE	0.2574348990962604	0.7968430648913267	0.016666666666666666	0.016666666666666666
2	BERTScore vs. SIDE	0.24790027320387326	0.8042115685178908	0.025	0.025
1	METEOR vs. BLEURT	0.03813850356981102	0.9695772522198308	0.05	0.05

Table 6: Holm / Shaffer Table for $\alpha = 0.10$

i	algorithms	$z = (R_0 - R_i)/SE$	p	Holm	Shaffer
55	IREs vs. InferSent	16.785708883668686	3.104995024128874E-63	0.0018181818181818182	0.0018181818181818182
54	BLEU-4 vs. IRES	14.573675676618908	4.130550871277846E-48	0.001851851851851852	0.0022222222222222222
53	ROUGE-L vs. IRES	12.986160465524987	1.4659915078915544E-38	0.0018867924528301887	0.0022222222222222222
52	IREs vs. SIDE	12.728725566428727	4.094681419615857E-37	0.0019230769230769232	0.0022222222222222222
51	BERTScore vs. IRES	12.480825293224854	9.499685886995614E-36	0.00196078431372549	0.0022222222222222222
50	SentenceBERT vs. InferSent	12.18525189055869	3.725006267054306E-34	0.002	0.0022222222222222222
49	IREs vs. SimLLM	10.392742222776953	2.675481714965443E-25	0.00204081633265306124	0.0022222222222222222
48	BLEU-4 vs. SentenceBERT	9.973218683508911	1.9965170154146324E-23	0.0020833333333333333	0.0022222222222222222
47	IREs vs. USE	9.71101647146638	2.7062466592891625E-22	0.002127659574468085	0.0022222222222222222
46	IREs vs. BLEURT	8.686044188027378	3.752789782724421E-18	0.002173913043478261	0.0022222222222222222
45	METEOR vs. IRES	8.647905684457566	5.245315319704807E-18	0.0022222222222222222	0.0022222222222222222
44	ROUGE-L vs. SentenceBERT	8.38570347241499	5.0423294553438346E-17	0.0022727272727272727	0.002702702702702703
43	METEOR vs. InferSent	8.13780319921112	4.0251441541170054E-16	0.002325581395348837	0.002702702702702703
42	SentenceBERT vs. SIDE	8.12826857331873	4.3546554116534997E-16	0.002380952380952381	0.002702702702702703
41	InferSent vs. BLEURT	8.09966469564131	5.511087099509436E-16	0.0024390243902439024	0.002702702702702703
40	BERTScore vs. SentenceBERT	7.880368300114856	3.264175041129758E-15	0.0025	0.002702702702702703
39	USE vs. InferSent	7.074692412202308	1.497806095635279E-12	0.002564102564102564	0.002702702702702703
38	InferSent vs. SimLLM	6.3929666608917355	1.6269774607603165E-10	0.002631578947368421	0.002702702702702703
37	METEOR vs. BLEU-4	5.9257699921613405	3.1083741177045455E-9	0.002702702702702703	0.002702702702702703
36	BLEU-4 vs. BLEURT	5.88763148859153	3.917694549050277E-9	0.0027777777777777778	0.0027777777777777778
35	SentenceBERT vs. SimLLM	5.792225229666955	6.943508882068498E-9	0.002857142857142857	0.0032258064516129032
34	SentenceBERT vs. USE	5.110559478356383	3.2120616484214425E-7	0.0029411764705882353	0.0032258064516129032
33	BLEU-4 vs. USE	4.862659205152527	1.158191566162712E-6	0.0030303030303030303	0.0032258064516129032
32	IREs vs. SentenceBERT	4.600456993109997	4.215650541578348E-6	0.003125	0.0032258064516129032
31	METEOR vs. ROUGE-L	4.338254781067421	1.4361860779334917E-5	0.0032258064516129032	0.0032258064516129032
30	BERTScore vs. InferSent	4.3048835904438345	1.6707356546645684E-5	0.0033333333333333335	0.003448275862068966
29	ROUGE-L vs. BLEURT	4.300116277497609	1.7070851629526506E-5	0.003448275862068966	0.003448275862068966
28	BLEU-4 vs. SimLLM	4.180933453841955	2.9031482616797085E-5	0.0035714285714285718	0.0035714285714285718
27	SentenceBERT vs. BLEURT	4.085587194917381	4.39654854323836E-5	0.0037037037037037034	0.0037037037037037034
26	METEOR vs. SIDE	4.08081988197116	4.4877121900045435E-5	0.0038461538461538464	0.004
25	InferSent vs. SIDE	4.056983317239961	4.9710631289064145E-5	0.004	0.004
24	METEOR vs. SentenceBERT	4.04744869134757	5.177892726152464E-5	0.004166666666666667	0.004166666666666667
23	BLEURT vs. SIDE	4.0426813784013484	5.2843406528614187E-5	0.004347826086956522	0.004347826086956522
22	METEOR vs. BERTScore	3.8329196087672863	1.266313451372739E-4	0.004545454545454546	0.004545454545454546
21	ROUGE-L vs. InferSent	3.7995484181437007	1.4495999007542706E-4	0.004761904761904762	0.004761904761904762
20	BERTScore vs. BLEURT	3.7947811051974756	1.477737716707003E-4	0.005	0.005
19	ROUGE-L vs. USE	3.2751439940586065	0.0010560812899933075	0.005263157894736842	0.005263157894736842
18	USE vs. SIDE	3.0177090949623464	0.002546932671489676	0.005555555555555556	0.005555555555555556
17	BERTScore vs. USE	2.769808821758473	0.0056089203911152	0.00588232941764705	0.00588232941764705
16	ROUGE-L vs. SimLLM	2.5934182427480343	0.009502712976774248	0.00625	0.00625
15	SIDE vs. SimLLM	2.3359833436517743	0.019492109652479973	0.006666666666666667	0.006666666666666667
14	BLEU-4 vs. InferSent	2.2120332070497803	0.026964370555333163	0.0071428571428571435	0.0071428571428571435
13	BLEU-4 vs. BERTScore	2.092850383394054	0.03636251257329595	0.007692307692307693	0.007692307692307693
12	BERTScore vs. SimLLM	2.088083070447901	0.03679034191083815	0.008333333333333333	0.008333333333333333
11	BLEU-4 vs. SIDE	1.8449501101901808	0.0650447962827122	0.009090909090909092	0.009090909090909092
10	METEOR vs. SimLLM	1.7448365383193858	0.08101332662327833	0.01	0.01
9	BLEURT vs. SimLLM	1.7066980347495746	0.08787819269001575	0.011111111111111112	0.011111111111111112
8	ROUGE-L vs. BLEU-4	1.5875152110939206	0.11239600465412686	0.0125	0.0125
7	METEOR vs. USE	1.0631107870088137	0.2877317131831188	0.014285714285714287	0.014285714285714287
6	USE vs. BLEURT	1.0249722834390025	0.3053762657842809	0.016666666666666666	0.016666666666666666
5	USE vs. SimLLM	0.6817257513105721	0.49541238110767266	0.02	0.02
4	ROUGE-L vs. BERTScore	0.5053351723001337	0.6133234430319932	0.025	0.025
3	ROUGE-L vs. SIDE	0.2574348990962604	0.7968430648913267	0.03333333333333333	0.03333333333333333
2	BERTScore vs. SIDE	0.24790027320387326	0.8042115685178908	0.05	0.05
1	METEOR vs. BLEURT	0.03813850356981102	0.9695772522198308	0.1	0.1

Table 7: Adjusted p -values

i	hypothesis	unadjusted p	P_{Neme}	P_{Holm}	P_{Shaf}	P_{Berg}
1	IREs vs .InferSent	.0000	.0000	.0000	.0000	.0000
2	BLEU-4 vs .IREs	.0000	.0000	.0000	.0000	.0000
3	ROUGE-L vs .IREs	.0000	.0000	.0000	.0000	.0000
4	IREs vs .SIDE	.0000	.0000	.0000	.0000	.0000
5	BERTScore vs .IREs	.0000	.0000	.0000	.0000	.0000
6	SentenceBERT vs .InferSent	.0000	.0000	.0000	.0000	.0000
7	IREs vs .SimLLM	.0000	.0000	.0000	.0000	.0000
8	BLEU-4 vs .SentenceBERT	.0000	.0000	.0000	.0000	.0000
9	IREs vs .USE	.0000	.0000	.0000	.0000	.0000
10	IREs vs .BLEURT	.0000	.0000	.0000	.0000	.0000
11	METEOR vs .IREs	.0000	.0000	.0000	.0000	.0000
12	ROUGE-L vs .SentenceBERT	.0000	.0000	.0000	.0000	.0000
13	METEOR vs .InferSent	.0000	.0000	.0000	.0000	.0000
14	SentenceBERT vs .SIDE	.0000	.0000	.0000	.0000	.0000
15	InferSent vs .BLEURT	.0000	.0000	.0000	.0000	.0000
16	BERTScore vs .SentenceBERT	.0000	.0000	.0000	.0000	.0000
17	USE vs .InferSent	.0000	.0000	.0000	.0000	.0000
18	InferSent vs .SimLLM	.0000	.0000	.0000	.0000	.0000
19	METEOR vs .BLEU-4	.0000	.0000	.0000	.0000	.0000
20	BLEU-4 vs .BLEURT	.0000	.0000	.0000	.0000	.0000
21	SentenceBERT vs .SimLLM	.0000	.0000	.0000	.0000	.0000
22	SentenceBERT vs .USE	.0000	.0000	.0000	.0000	.0000
23	BLEU-4 vs .USE	.0000	.0001	.0000	.0000	.0000
24	IREs vs .SentenceBERT	.0000	.0002	.0001	.0001	.0000
25	METEOR vs .ROUGE-L	.0000	.0008	.0004	.0004	.0000
26	BERTScore vs .InferSent	.0000	.0009	.0005	.0005	.0000
27	ROUGE-L vs .BLEURT	.0000	.0009	.0005	.0005	.0000
28	BLEU-4 vs .SimLLM	.0000	.0016	.0008	.0008	.0000
29	SentenceBERT vs .BLEURT	.0000	.0024	.0012	.0012	.0000
30	METEOR vs .SIDE	.0000	.0025	.0012	.0012	.0000
31	InferSent vs .SIDE	.0000	.0027	.0012	.0012	.0000
32	METEOR vs .SentenceBERT	.0001	.0028	.0012	.0012	.0000
33	BLEURT vs .SIDE	.0001	.0029	.0012	.0012	.0000
34	METEOR vs .BERTScore	.0001	.0070	.0028	.0028	.0000
35	ROUGE-L vs .InferSent	.0001	.0080	.0030	.0030	.0000
36	BERTScore vs .BLEURT	.0001	.0081	.0030	.0030	.0000
37	ROUGE-L vs .USE	.0011	.0581	.0201	.0201	.0000
38	USE vs .SIDE	.0025	.1401	.0458	.0458	.0000
39	BERTScore vs .USE	.0056	.3085	.0954	.0954	.0000
40	ROUGE-L vs .SimLLM	.0095	.5226	.1520	.1520	.0000
41	SIDE vs .SimLLM	.0195	1.0721	.2924	.2924	.0000
42	BLEU-4 vs .InferSent	.0270	1.4830	.3775	.3775	.0000
43	BLEU-4 vs .BERTScore	.0364	1.9999	.4727	.4727	.0000
44	BERTScore vs .SimLLM	.0368	2.0235	.4727	.4727	.0000
45	BLEU-4 vs .SIDE	.0650	3.5775	.7155	.7155	.0000
46	METEOR vs .SimLLM	.0810	4.4557	.8101	.8101	.0000
47	BLEURT vs .SimLLM	.0879	4.8333	.8101	.8101	.0000
48	ROUGE-L vs .BLEU-4	.1124	6.1818	.8992	.8992	.0000
49	METEOR vs .USE	.2877	15.8252	2.0141	2.0141	.0000
50	USE vs .BLEURT	.3054	16.7957	2.0141	2.0141	.0000
51	USE vs .SimLLM	.4954	27.2477	2.4771	2.4771	.0000
52	ROUGE-L vs .BERTScore	.6133	33.7328	2.4771	2.4771	.0000
53	ROUGE-L vs .SIDE	.7968	43.8264	2.4771	2.4771	.0000
54	BERTScore vs .SIDE	.8042	44.2316	2.4771	2.4771	.0000
55	METEOR vs .BLEURT	.9696	53.3267	2.4771	2.4771	.0000